Nano-Sensors (100pk) - User Manual

Official Technical Specification, Operating Instructions & Support Policy

Category:	Electronics	**Retail Price:**	$89.99
Model Number:	NS-100-MEMS	**Warranty Term:**	1 Month Limited

1. Product Overview & Technical Specifications

The Nano-Sensors (100pk) package delivers an industrial-grade array of one hundred miniaturized micro-electromechanical (MEMS) multi-sensor nodes. Engineered for comprehensive environmental monitoring, predictive equipment maintenance, structural health assessment, and distributed agricultural sensing, each thumbnail-sized node integrates precision sensors for temperature (-40°C to +125°C), relative humidity (0-100% RH), tri-axial acceleration, and localized magnetic flux. Nodes communicate via an ultra-low power Sub-GHz mesh wireless protocol capable of line-of-sight transmission up to 50 meters. Encased in IP67 water- and dust-resistant epoxy resin with pre-applied industrial thermal adhesive backing, each sensor incorporates an ambient energy-harvesting cell supported by a 3-year micro-battery reserve.

2. Setup & Operating Instructions

Deploying the Nano-Sensors array across your target monitoring zone is efficient and scalable. First, set up and power on the central Nano-Gateway receiver connected to your local network. Unpack the required number of sensor nodes from the anti-static packaging. Remove the pull-tab plastic insulator from the rear of each sensor to engage internal circuitry; a faint green micro-LED will flash once. Open the NanoMesh Gateway dashboard on your mobile device or terminal, and scan the individual 2D DataMatrix code on each node to register its unique telemetry ID and location tag. Peel off the protective adhesive film on the base and press the node firmly against the test surface for 10 seconds to ensure solid mechanical coupling for vibration sensing and thermal contact.

3. Troubleshooting & Diagnostics Guide

In case of communication or telemetry irregularities, refer to the following guide. Issue 1: A registered node ceases transmitting telemetry to the gateway. Resolution: Check if the distance between the node and the nearest mesh neighbor exceeds 50 meters; install an intermediate repeater node or relocate the gateway to overcome thick concrete shielding. Issue 2: Sensor outputs irregular temperature or humidity spikes. Resolution: Inspect the protective membrane on the node aperture for oil, grease, or particulate clogging; clean gently using isopropyl alcohol and a lint-free swab. Issue 3: Gateway fails to recognize scanned node QR code. Resolution: Manually enter the 12-digit hex MAC ID printed beneath the barcode into the gateway's manual device registration portal.

4. Warranty Terms & Customer Support Information

Warranty Summary: 1 Month (30 Days) Limited Manufacturer Protection

The Nano-Sensors (100pk) is protected by a 1-month (30-day) limited replacement warranty covering factory component defects and dead-on-arrival (DOA) nodes. If any individual sensor node fails to initialize, negotiate mesh connectivity, or register accurate telemetry within 30 days of purchase, it will be replaced promptly upon verification. This 1-month warranty does not cover nodes subjected to mechanical crushing beyond 500 N, immersion in caustic chemical solvents, or intentional encapsulation stripping. For warranty claims, email support@agentichelpdesk.com with the affected sensor MAC IDs and a screenshot of the gateway's offline device log to receive replacement sensor hardware. For technical assistance scheduled firmware maintenance compliance records and certified servicing documentation for the Nano-Sensors 100pk customers may.